4.3.1. Commissionable Node Discovery

The Matter protocol family supports UDP and TCP for Matter commissioning of Commissionees already on the customer's IP network (such as Ethernet-connected Nodes, or Wi-Fi Nodes already associated to the Wi-Fi network via other means).

For these Commissionees, Matter Commissionable Node Discovery is performed using IETF Standard DNS-Based Service Discovery (DNS-SD) [RFC 6763] as described below.

For Matter Commissionable Node Discovery in the *already-on-network* case, the DNS-SD instance name SHALL be a dynamic, pseudo-randomly selected, 64-bit temporary unique identifier, expressed as a fixed-length sixteen-character hexadecimal string, encoded as ASCII (UTF-8) text using capital letters, e.g., `DD200C20D25AE5F7`. A new instance name SHALL be selected when the Node boots. A new instance name SHALL be selected whenever the Node enters Commissioning mode. A new instance name MAY be selected at other times, as long as the instance name does not change while the Node is in commissioning mode.

A commissionable Node that is already connected to an IP-bearing network SHALL only make itself discoverable on the IP network and SHALL use the relevant DNS-SD service (`_matterc._udp`) described below.

If a Node is connected to multiple IP-bearing networks, and it receives either the OpenCommissioningWindow or the OpenBasicCommissioningWindow command, it MAY make itself discoverable only on the network on which the command was received. Otherwise, a commissionable Node that is connected to multiple IP-bearing networks SHALL make itself discoverable on all of its connected IP-bearing networks.

The Matter Commissionable Node Discovery DNS-SD instance name SHALL be unique within the namespace of the local network (the `.local` link-local namespace of the Ethernet and Wi-Fi links [RFC 6762], or the unicast domain selected by the Thread Border Router for devices on the Thread mesh).

In the rare event of a collision in the selection of the 64-bit temporary unique identifier, the existing DNS-SD name conflict detection mechanism will detect this collision, and a new pseudo-randomly selected 64-bit temporary unique identifier SHALL be generated by the Matter Commissionee that is preparing for commissioning. Name conflict detection is described in Section 9 ("Conflict Resolution") of the Multicast DNS specification [RFC 6762] and Section 2.4.3.1 ("Validation of Adds") of the Service Registration Protocol specification [SRP].

The DNS-SD service type [RFC 6335] for Matter Commissionable Node Discovery is `_matterc._udp`.

For link-local Multicast DNS the service domain SHALL be `local`. For Unicast DNS such as used on Thread the service domain SHALL be as configured automatically by the Thread Border Router.

4.3.1.1. Host Name Construction

For DNS-SD a target host name is required, in addition to the instance name. The target host name SHALL be constructed using one of the available link-layer addresses, such as a 48-bit device MAC address (for Ethernet and Wi-Fi) or a 64-bit MAC Extended Address (for Thread) expressed as a fixed-length twelve-character (or sixteen-character) hexadecimal string, encoded as ASCII (UTF-8)

